

Ex. No 1: Evidence Acquisition Using AccessData FTK Imager

Course / Lab: Digital Forensics Laboratory

Experiment: Evidence Acquisition Using AccessData FTK Imager[cite: 2]

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Overview

Forensic Toolkit (FTK) Imager is an industry-standard computer forensics software product used for acquiring and analyzing digital evidence securely[cite: 2]. This experiment demonstrates the methodology for acquiring both volatile memory (RAM) and non-volatile storage (logical partitions) while preserving evidence integrity through cryptographic hashing.

Experiment Objectives

1. Capture volatile memory and system states from a live environment.
 2. Acquire non-volatile evidence via logical drive/partition imaging to optimize laboratory time[cite: 3].
 3. Verify data integrity by computing and validating MD5 and SHA1 checksums[cite: 3].
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Part 1: Acquiring Volatile Memory (RAM)

Volatile memory captures running processes, active network sockets, and temporary data.

Steps:

1. Launch the memory acquisition utility (e.g., WinPmem/FTK memory capture module) with administrator privileges[cite: 3].
2. Define the output destination file (.raw / .mem) and monitor the live memory dumping progress[cite: 3].
3. Allow the driver to complete the memory transfer and verify driver unloading.

Capturing Memory in Progress

[image]

Memory Capture Completed

[image]

Figure 1.1: Live memory dumping execution. Figure 1.2: Successful memory dump completion.

Part 2: Acquiring Non-Volatile Memory (Logical Drive / Partition)

To streamline forensic acquisition workflows, a specific partition (F:\) is targeted instead of

the entire physical disk.

Steps:

1. Open FTK Imager and initiate the **Create Disk Image** wizard[cite: 3].
2. Select **Logical Drive** as the source type and choose the target partition (F : \)[cite: 3].
3. Fill in the **Evidence Item Information** (Case Number, Evidence Number, Unique Description, Examiner, and Notes)[cite: 3].
4. Configure the destination folder (E : \), image filename (EXP), fragment size, and enable **Verify images after they are created**[cite: 3].
5. Click **Start** to begin imaging[cite: 3].

Evidence Item Information Image Source & Destination Setup

[image]

Figure 2.1: Case and examiner metadata.

[image]

Figure 2.2: Source partition selection (F : \).

Configuring Destination & Fragment Size

[image]

Figure 2.3: Destination path and options.

🔍 Verification & Cryptographic Hashes

To ensure court admissibility and data integrity, FTK Imager automatically calculates cryptographic hashes upon completion and cross-verifies them against the report hash[cite: 3].

Verification Result

[image]

Figure 3.1: Drive/Image Verify Results showing successful MD5 and SHA1 matches.

Checksum Summary:

Image Type: Raw / Logical Drive Image (Result.001)
Sector Count: 31506432

[MD5 Hash]

Computed hash: 1f8957e55573455d076e144a74afef4
Report hash: 1f8957e55573455d076e144a74afef4
Verify result: Match

[SHA1 Hash]

Computed hash: 613f621a072747853feb89ed81700c749c2cc7b1
Report hash: 613f621a072747853feb89ed81700c749c2cc7b1
Verify result: Match

[Bad Blocks List]

Bad block(s) in image: No bad blocks found in image